



Strategi Data Sourcing

Spot Istimewa — Gaskuy

Panduan lengkap mengumpulkan & mengelola data destinasi roadtrip Indonesia
dengan metode 100% gratis.

Versi: 1.0 • **Tanggal:** Mei 2026

Proyek: Gaskuy — Platform Roadtrip Indonesia

Status: MVP (25 spots manual)

Daftar Isi

1 Ringkasan Eksekutif

2 Arsitektur Data

3 Metode Manual (Current)

4 Google Form Submission

5 OpenStreetMap Overpass API

6 Wikipedia API

7 Web Scraping (Opsional)

8 Pipeline Automation

9 Implementation Roadmap

10 Maintenance Guide

A Appendix — Data Schema

B Appendix — Province-to-City Mapping

C Appendix — API Quotas & Limits

1. Ringkasan Eksekutif

Halaman **/spot-istimewa** adalah fitur kurasi destinasi roadtrip Indonesia. Saat ini berisi **25 spots** yang diinput secara manual. Untuk scale ke ratusan hingga ribuan spots, diperlukan pipeline data yang sistematis.

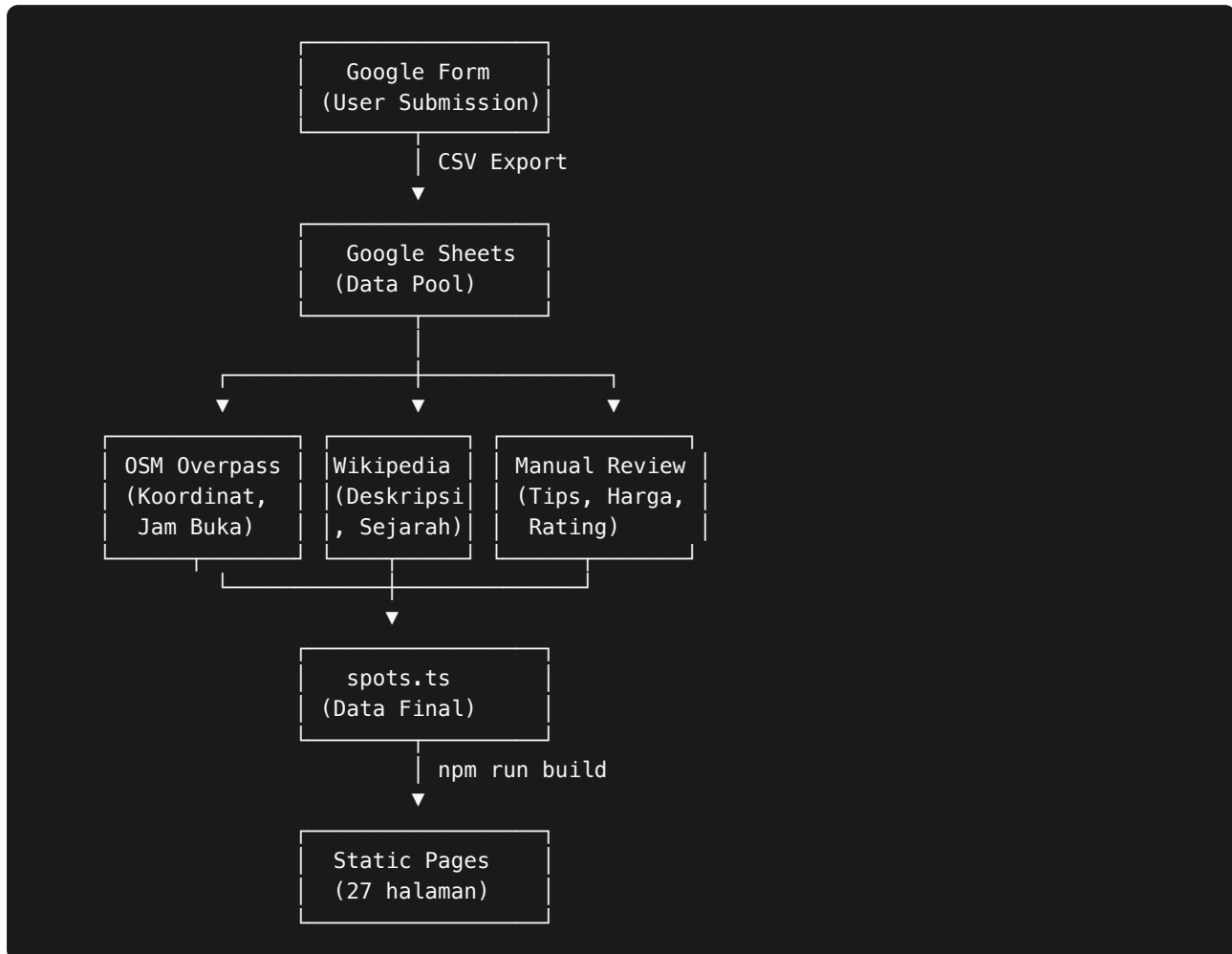
Dokumen ini merangkum 4 metode pengumpulan data gratis yang bisa digunakan:

Metode	Effort	Data Quality	Scalability	Best Untuk
Manual Curation	30 menit/spot	★★★★★	✗	MVP, featured spots
Google Form	30 menit setup	★★★	✓	Community submissions
OpenStreetMap API	2-3 hari setup	★★★	✓✓	Bulk POI (koordinat, jam)
Wikipedia API	1 hari setup	★★★★	✓	Deskripsi, sejarah, fakta unik

Strategi: Manual curation untuk 50 spots featured + Google Form untuk community submissions. Enrich data dengan OSM & Wikipedia API untuk auto-populate field.

2. Arsitektur Data

2.1 Data Flow



2.2 Data Schema

```

Spot {
  slug: string           // URL slug (auto-generate dari nama)
  name: string            // Nama tempat
  category: SpotCategory // Alam | Kuliner | Budaya | Foto | Petualangan | Sejarah
  province: string        // Provinsi
  region: string          // Region: Jawa, Sumatera, dll
  location: {             // Koordinat (OSM API)
    lat: number
    lng: number
  }
  description: string     // Deskripsi (Wikipedia + Manual)
  whySpecial: string      // Keunikan (Manual)
  rating: number          // Rating (OSM + Manual)
  imageUrl: string        // Foto (Download + Local)
  tags: string[]          // Tags (Auto dari kategori + manual)
}
  
```

```
tips: string           // Tips (Manual)
bestTime: string       // Waktu terbaik (Manual)
openingHours: string   // Jam buka (OSM API)
estimatedTime: string  // Estimasi waktu explore (Manual)
ticketPrice: string    // Harga tiket (Manual)
roadAccess: string     // Akses jalan (Manual)
facilities: string[]   // Fasilitas (OSM API)
distanceFromCity: string // Jarak dari kota (Manual)
}
```

Catatan: Field yang bisa dipopulate otomatis = location, openingHours, facilities, tags.
Sisanya butuh review manual.

3. Metode Manual (Current)

Metode yang dipakai untuk 25 spots saat ini. Ideal untuk featured spots berkualitas tinggi.

3.1 Workflow per Spot

1. **Riset:** Google Maps + blog travel Indonesia (jalantikus, pergi.com, traveloka blog)
2. **Verifikasi:** Cek jam buka, harga, koordinat di Google Maps
3. **Konten:** Tulis deskripsi, tips, whySpecial (3-5 kalimat)
4. **Foto:** Download dari Unsplash (resolusi 800px, simpan di `/public/images/spots/`)
5. **Input:** Tambah data ke `src/data/spots.ts`
6. **Build:** `npm run build` untuk generate halaman statis

3.2 Template Deskripsi

```
deskripsi: "Kalimat 1: Lokasi + daya tarik utama.  
          Kalimat 2: Detail tambahan (suasana, aktivitas, keunikan)."  
  
whySpecial: "Kenapa ini wajib dikunjungi?  
            Fakta unik atau record yang membedakan dari tempat lain."
```

3.3 Sumber Riset Gratis

Sumber	Data	Akses
Google Maps	Jam buka, harga, rating, foto, review	gratis (manual)
Wikipedia	Deskripsi, sejarah, koordinat	gratis (API)
Blog Travel	Tips, pengalaman, harga terkini	gratis (manual)
Instagram	Tags, hashtag, foto terkini	gratis (manual)
Google Trends	Popularitas destinasi	gratis

4. Google Form Submission

4.1 Setup

1. Buka forms.google.com
2. Buat form baru: "**Submit Spot Istimewa**"
3. Tambah fields berikut:

4.2 Fields Google Form

Field	Tipe	Required	Validasi
Nama Tempat	Short text	✓	-
Provinsi	Dropdown	✓	34 provinsi Indonesia
Kategori	Dropdown	✓	Alam, Kuliner, Budaya, Foto, Petualangan, Sejarah
Link Google Maps	URL	✓	Valid URL format
Deskripsi Singkat	Paragraph	✓	Min 50 karakter
Kenapa Istimewa?	Paragraph	-	Min 30 karakter
Tips	Paragraph	-	-
Harga Tiket	Short text	-	-
Jam Buka	Short text	-	-
Link Foto	URL	-	Optional, bisa upload sendiri
Nama Kamu	Short text	✓	Untuk kredit
Email	Email	✓	Valid email

4.3 Workflow Moderation

1. User submit → Google Sheets
2. Kamu review (weekly)
3. Cek validitas data (Google Maps)
4. Enrich dengan OSM API
5. Tulis deskripsi final
6. Download foto
7. Merge ke spots.ts
8. Build & deploy

Estimasi waktu review: 5-10 menit per submission (termasuk verifikasi + enrichment).

5. OpenStreetMap Overpass API

5.1 Overview

OpenStreetMap (OSM) adalah database geografis gratis dan open-source. Overpass API memungkinkan query data POI secara terstruktur. Cocok untuk mengambil data koordinat, jam buka, fasilitas, dan rating.

5.2 Query Examples

Query: Cari SPBU di Bandung

```
// Overpass QL
[out:json];
node
  ["amenity"="fuel"]
  ["name"~"pertamina",i]
  (area:searchArea);
out body 10;
```

Query: Cari Restoran di Yogyakarta

```
[out:json];
node
  ["amenity"="restaurant"]
  (area:searchArea);
out body 20;
```

Query: Cari Tempat Wisata Alam di Jawa Barat

```
[out:json];
(
  node["tourism"="attraction"](area:searchArea);
  node["leisure"="nature_reserve"](area:searchArea);
  node["waterway"="waterfall"](area:searchArea);
);
out body 15;
```

5.3 Python Script: Enrichment Pipeline

```
import requests, json

OVERPASS_URL = "https://overpass-api.de/api/interpreter"
```

```
def enrich_spot(name: str, lat: float, lng: float) -> dict:
    """Enrich spot data from OSM."""
    query = f"""
    [out:json];
    (
        node(around:100,{lat},{lng});
        way(around:100,{lat},{lng});
    );
    out body 5;
    """
    resp = requests.get(OVERPASS_URL, params={"data": query},
                        timeout=15)

    # Parse response
    elements = resp.json().get("elements", [])
    if not elements:
        return {}

    tags = elements[0].get("tags", {})
    return {
        "opening_hours": tags.get("opening_hours", ""),
        "phone": tags.get("phone", ""),
        "website": tags.get("website", ""),
        "wheelchair": tags.get("wheelchair", ""),
        "toilets": tags.get("toilets", ""),
    }

# Example usage
result = enrich_spot("Kawah Putih", -7.1660, 107.4042)
print(result)
```

5.4 Python Script: Bulk Search by Category

```
import requests, time

def search_poi_by_category(category: str, lat: float, lng: float,
                           radius: int = 50000):
    """Search POI by category around a location."""
    osm_tags = {
        "kuliner": '["amenity"="restaurant"]',
        "alam": '["tourism"="attraction"] ["natural"]',
        "budaya": '["tourism"="museum"]',
        "spbu": '["amenity"="fuel"]',
        "bengkel": '["amenity"="car_repair"]',
    }
    tag_query = osm_tags.get(category, '["tourism"="attraction"]')

    query = f"""
    [out:json];
    node{tag_query}(around:{radius},{lat},{lng});
    out body 20;
    """
    resp = requests.get("https://overpass-api.de/api/interpreter",
                        params={"data": query})
    time.sleep(2) # Rate limiting
    return resp.json().get("elements", [])

# Example: cari kuliner di sekitar Borobudur
```

```
spots = search_poi_by_category("kuliner", -7.6079, 110.2038)
for s in spots[:5]:
    print(f"{s.get('tags', {}).get('name', '-')}")
```

Rate Limit: Overpass API gratis, maksimal 10 request per menit. Gunakan caching untuk production.

6. Wikipedia API

6.1 Overview

Wikipedia API gratis, tanpa rate limit ketat. Cocok untuk mengambil deskripsi, sejarah, dan fakta unik dari spot.

6.2 Python Script: Fetch Description

```
import requests, urllib.parse

WIKI_API = "https://id.wikipedia.org/w/api.php"
# Gunakan bahasa Indonesia untuk spot Indonesia

def fetch_wikipedia_description(spot_name: str) -> dict:
    """Fetch description from Wikipedia Indonesia."""
    params = {
        "action": "query",
        "format": "json",
        "titles": spot_name,
        "prop": "extracts|coordinates|pageimages",
        "exintro": True,
        "explaintext": True,
        "exsentences": 5,
        "pithumbsize": 500,
    }
    resp = requests.get(WIKI_API, params=params, timeout=10)
    data = resp.json()
    pages = data.get("query", {}).get("pages", {})
    for page_id, page in pages.items():
        if page_id == "-1":
            continue # Page not found
        return {
            "description": page.get("extract", ""),
            "coordinates": page.get("coordinates", [{}])[0]
                           if page.get("coordinates") else None,
            "thumbnail": page.get("thumbnail", {}).get("source", ""),
        }
    return {}

# Example: ambil deskripsi Borobudur
result = fetch_wikipedia_description("Candi Borobudur")
print(result["description"][:200])
```

6.3 Extract "Why Special" dari Wikipedia

```
def extract_why_special(description: str) -> str:
    """Extract key facts from Wikipedia description."""
    import re
    # Cari kalimat dengan kata kunci
```

```
keywords = ["terbesar", "tertinggi", "tertua", "satu-satunya",  
            "terbaik", "terkenal", "ikon", "warisan dunia",  
            "terindah", "legendaris"]  
sentences = re.split(r'(?<=[.!?])\s+', description)  
special = []  
for s in sentences:  
    for kw in keywords:  
        if kw in s.lower() and len(s) > 20:  
            special.append(s.strip())  
            break  
return " ".join(special[:3]) if special else description[:200]
```

7. Web Scraping (Opsional)

⚠️ **Legal Disclaimer:** Scraping website bisa melanggar Terms of Service. Lakukan dengan bijak dan hormati robots.txt. Gunakan hanya untuk data publik dan rate limit yang wajar.

7.1 Target Websites

Website	Data	Risiko	Notes
Traveloka Blog	Destinasi, tips	Rendah	Content blog publik
Pergi.com	Destinasi, harga	Rendah	Public info
Wikipedia	Deskripsi	Tidak ada	API resmi tersedia ✅
Google Maps	Rating, jam buka	Tinggi	Gunakan Places API
Instagram	Foto, tags	Tinggi	ToS violation risk

7.2 Scraping dengan Python

```
import requests
from bs4 import BeautifulSoup

def scrape_blog_articles(url: str) -> list:
    """Scrape travel blog articles."""
    headers = {'User-Agent': 'Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)'}
    resp = requests.get(url, headers=headers, timeout=10)
    soup = BeautifulSoup(resp.text, 'html.parser')

    articles = []
    for article in soup.select('article')[:5]:
        title = article.select_one('h2, h3')
        desc = article.select_one('p')
        if title:
            articles.append({
                'title': title.get_text(strip=True),
                'snippet': desc.get_text(strip=True) if desc else '',
            })
    return articles
```

7.3 Best Practices

- **Rate limit:** Max 1 request per 3 detik
- **User-Agent:** Gunakan identitas yang jelas

- **Robots.txt:** Cek `/robots.txt` sebelum scraping
- **Cache:** Simpan hasil scraping, jangan scrape ulang
- **Attribution:** Kredit sumber jika diperlukan

8. Pipeline Automation

8.1 Full Enrichment Script

```
import requests, json, time, re

# --- CONFIG ---
OVERPASS_URL = "https://overpass-api.de/api/interpreter"
WIKI_API = "https://id.wikipedia.org/w/api.php"

def enrich_spot(name: str, lat: float, lng: float) -> dict:
    """Full enrichment pipeline."""
    result = {"name": name, "lat": lat, "lng": lng}

    # 1. OSM Enrichment
    query = f"[out:json];node(around:50,{lat},{lng});out body 5;"
    try:
        resp = requests.get(OVERPASS_URL, params={"data": query},
                             timeout=10)
        elements = resp.json().get("elements", [])
        if elements:
            tags = elements[0].get("tags", {})
            result["opening_hours"] = tags.get("opening_hours", "")
            result["phone"] = tags.get("phone", "")
            result["website"] = tags.get("website", "")
    except Exception as e:
        print(f" OSM error: {e}")

    time.sleep(2) # Rate limit

    # 2. Wikipedia Enrichment
    params = {
        "action": "query", "format": "json",
        "titles": name, "prop": "extracts|coordinates",
        "exintro": True, "explaintext": True, "exsentences": 5,
    }
    try:
        resp = requests.get(WIKI_API, params=params, timeout=10)
        pages = resp.json().get("query", {}).get("pages", {})
        for pid, page in pages.items():
            if pid != "-1":
                result["wikipedia_desc"] = page.get("extract", "")
                coords = page.get("coordinates", [{}])[0]
                if coords:
                    result["wiki_lat"] = coords.get("lat")
                    result["wiki_lng"] = coords.get("lon")
    except Exception as e:
        print(f" Wiki error: {e}")

    return result

# --- BATCH PROCESS ---
SPOTS = [
    {"name": "Kawah Putih", "lat": -7.1660, "lng": 107.4042},
    {"name": "Candi Borobudur", "lat": -7.6079, "lng": 110.2038},
    {"name": "Gunung Bromo", "lat": -7.9425, "lng": 112.9530},
```



```
]

for spot in SPOTS:
    print(f"Enriching: {spot['name']}...")
    data = enrich_spot(spot["name"], spot["lat"], spot["lng"])
    print(json.dumps(data, indent=2))
    print("----")
```

8.2 Automation Workflow

Weekly Batch Script

1. Export Google Form → CSV
2. Run Python enrichment script
3. Manual review & edit
4. Generate draft spots.ts
5. npm run build & verify
6. Manual git commit

Weekly schedule: Setiap Senin pagi, batch process submission minggu sebelumnya.
Estimasi: 30-60 menit total.

9. Implementation Roadmap

Phase 1: Launch (Sekarang)

Task	Status	Effort
25 spots manual curation	 Done	1 minggu
Halaman listing + detail	 Done	2 hari
Region filter	 Done	-
Rute Populer integration	 Done	-

Phase 2: Post-Launch (Minggu 2-4)

Task	Effort	Priority
Google Form submission	30 menit	 High
Add "Submit Spot" link di website	30 menit	 High
Python enrichment script	2 hari	 Medium
Tambah 25 spots baru	1 hari	 Medium
Foto spesifik per spot (bukan generic)	2 hari	 Medium

Phase 3: Scale (Bulan 2+)

Task	Effort	Priority
Setup Supabase + backend database	1 minggu	 High
Build submission form di app (no Google Form)	3 hari	 Medium
Admin moderation panel	1 minggu	 Medium
Auto-approve for trusted users	2 hari	 Low
100+ spots in database	2 minggu	 High

10. Maintenance Guide

10.1 Weekly Workflow Checklist

```
[ ] Export Google Form submissions from Sheets
[ ] Validate: cek koordinat, jam buka, harga via Google Maps
[ ] Run Python enrichment script (OSM + Wikipedia)
[ ] Tulis deskripsi final (1-2 paragraf)
[ ] Download foto dari Unsplash (800px, landscape)
[ ] Update src/data/spots.ts
[ ] npm run build
[ ] Verify: buka localhost, cek 3-5 halaman random
[ ] Git commit & push
[ ] Deploy
```

10.2 Quality Control Checklist

- ☒ Nama tempat sesuai ejaan yang benar
- ☒ Koordinat akurat (cek di Google Maps)
- ☒ Deskripsi minimal 50 karakter, informatif
- ☒ Tips praktis, bukan generic
- ☒ Harga tiket update (cek bulan ini)
- ☒ Jam buka valid (cek Google Maps)
- ☒ Foto tidak pecah/blur
- ☒ Slug unik, tanpa duplikasi
- ☒ Build success (36+ pages)

10.3 Troubleshooting

Masalah	Cause	Solution
Image not found (404)	File tidak ada di /public/images/spots/	Cek path, download ulang dari Unsplash
Build error: duplicate key	Slug tidak unique	Gunakan slug generator: nama + lokasi
OSM API timeout	Rate limit exceeded	Tambahkan sleep(3) antar request
Wikipedia page not found	Judul berbeda di Wikipedia	Coba alternatif: "Kawah Putih" → "Kawah Putih Ciwidey"

Foto terlalu besar	Resolusi > 800px	Resize: convert -resize 800x input.jpg output.jpg
--------------------	------------------	--

Lampiran A: Data Schema Lengkap

```
interface Spot {
  slug: string
  name: string
  category: "alam" | "kuliner" | "budaya" | "foto" | "petualangan" | "sejarah"
  province: string
  region: string
  location: { lat: number; lng: number }
  description: string
  whySpecial: string
  rating: number
  imageUrl: string
  imageCredit: string
  tags: string[]
  addedAt: Date
  tips: string
  bestTime: string
  openingHours: string
  estimatedTime: string
  ticketPrice: string
  roadAccess: string
  facilities: string[]
  distanceFromCity: string
  popularRoutes?: { from: string; duration: string }[]
  relatedProductIds?: string[]
}

type SpotCategory = "alam" | "kuliner" | "budaya" | "foto" | "petualangan" | "sejarah"

// Category config
{
  alam: { label: "Alam", icon: "🌄", color: "emerald" }
  kuliner: { label: "Kuliner", icon: "🍲", color: "orange" }
  budaya: { label: "Budaya", icon: "🏛️", color: "purple" }
  foto: { label: "Spot Foto", icon: "📷", color: "pink" }
  petualangan: { label: "Petualangan", icon: "🗺️", color: "blue" }
  sejarah: { label: "Sejarah", icon: "📖", color: "amber" }
}
```

Lampiran B: Province-to-City Mapping

```
const PROVINCE_ROUTES = {  
  "Jakarta": ["Jakarta", "Bandung"],  
  "Jawa Barat": ["Jakarta", "Bandung", "Bogor"],  
  "Jawa Tengah": ["Yogyakarta", "Semarang", "Solo"],  
  "DIY": ["Yogyakarta", "Semarang", "Solo"],  
  "Jawa Timur": ["Surabaya", "Malang"],  
  "Bali": ["Denpasar", "Canggu", "Kuta"],  
  "NTB": ["Mataram", "Lombok Airport"],  
  "NTT": ["Labuan Bajo"],  
  "Sumatera Utara": ["Medan"],  
  "Sumatera Barat": ["Padang", "Medan"],  
  "Sulawesi Utara": ["Manado"],  
  "Sulawesi Selatan": ["Makassar"],  
  "Kalimantan Timur": ["Balikpapan", "Tanjung Redeb"],  
}
```

Lampiran C: API Quotas & Limits

API	Free Quota	Rate Limit	Notes
Overpass API	Unlimited	~10 req/min	Turun drastis kalo overload
Wikipedia API	Unlimited	~200 req/min	Fair use, jangan abuse
Google Places API	\$200 credit/bln	~180 req/min	Pilih yang gratis dulu
Unsplash API	50 req/jam	50 req/jam	Hanya untuk search, download unlimited
OpenStreetMap Nominatim	Unlimited	1 req/dtk	Gunakan User-Agent yang jelas

Dokumen ini dibuat untuk proyek Gaskuy • Spot Istimewa • 2026